AIP | Journal of Mathematical Physics

The N -vortex problem on a symmetric ellipsoid: A perturbation approach

César Castilho and Hélio Machado

Citation: Journal of Mathematical Physics **49**, 022703 (2008); doi: 10.1063/1.2863515
View online: http://dx.doi.org/10.1063/1.2863515
View Table of Contents: http://scitation.aip.org/content/aip/journal/jmp/49/2?ver=pdfcov
Published by the AIP Publishing

JOURNAL OF MATHEMATICAL PHYSICS **49**, 022703 (2008)

The *N*-vortex problem on a symmetric ellipsoid: A perturbation approach

César Castilho[1,2,a)] and Hélio Machado[1]
[1]*Departamento de Matemática, Universidade Federal de Pernambuco, Recife, Pernambuco CEP 50740-540, Brazil*
[2]*The Abdus Salam ICTP, Strada Costiera 11, Trieste 34100, Italy*

(Received 16 October 2007; accepted 23 January 2008; published online 26 February 2008)

We consider the N-vortex problem on an ellipsoid of revolution. Applying standard techniques of classical perturbation theory, we construct a sequence of conformal transformations from the ellipsoid into the complex plane. Using these transformations, the equations of motion for the N-vortex problem on the ellipsoid are written as a formal series on the eccentricity of the ellipsoid's generating ellipse. First order equations are obtained explicitly. We show numerically that the truncated first order system for the three vortex system on the symmetric ellipsoid is nonintegrable.
 [DOI: 10.1063/1.2863515]

I. INTRODUCTION

The equations of point vortex motion in the plane were introduced by Helmholtz[1] and described as a Hamiltonian system by Kirchhoff in 1876.[2] These equations were first generalized to describe the motion of point vortices on a sphere by Bogolmonov in 1977.[3] Bogomolnov's work was put into solid mathematical grounds by Kimura and Okamoto in 1987.[4] Independently, Hally wrote in 1979 (Ref. 5) the equations for vortex motion in symmetric surfaces of revolution (which englobe the sphere). To write his equations, Hally represented the surface M on the complex plane with a metric conformal to the Euclidean metric: Let the ith vortex, with vorticity Γ_i, be represented by the complex coordinate ξ_i. Calling the surface's conformal factor $h(\xi,\bar{\xi})$, Hally's equations are

$$\bar{\xi}_n' = h^{-2}(\xi_n,\bar{\xi}_n)\left(\sum_k{}' - i\frac{\Gamma_k}{\xi_n - \xi_k} + i\Gamma_n\frac{\partial}{\partial \xi_n}\log(h(\xi_n,\bar{\xi}_n))\right). \tag{1}$$

The difficulty in working with these equations is that one must know explicitly the conformal factor $h(\xi,\bar{\xi})$ and this factor is not known for many surfaces but the sphere.[6–8]

The aim of this paper is to write the equations for the N-vortex problem on the ellipsoid of revolution

$$\frac{x^2}{R^2} + \frac{y^2}{R^2} + \frac{z^2}{R^2(1+\epsilon)} = 1,$$

as an ϵ-series when $|\epsilon| \ll 1$. To achieve this goal, we write a perturbative coordinate transformation from the ellipsoid into the plane [see (16)] and impose conformality at each order. It turns out that the transformation's nth order term is given by the solution of a certain linear differential equation and that the first order term can be computed explicitly [see (17)]. Using Hally's equations, we calculated rigorously the first order perturbation term for the vortex equations on the ellipsoid. In spherical coordinates, the truncated first order equations are given by the Hamiltonian system with Hamiltonian

a)Electronic mail: castilho@dmat.ufpe.br.

$$H = H_0 + \epsilon H_1, \tag{2}$$

where

$$H_0 = \frac{1}{2}\sum_{i,j}{}' \Gamma_i\Gamma_j[\ln(2 - 2\cos(\theta_i)\cos(\theta_j) - 2\sin(\theta_i)\sin(\theta_j)\cos(\phi_i - \phi_j))],$$

$$H_1 = \frac{1}{2}\sum_{i,j}{}' \Gamma_i\Gamma_j[(\cos(\theta_i)^2 + \cos(\theta_j)^2)],$$

and symplectic 2-form

$$w = \sum_{i=1}^{N} \Gamma_i \sin(\theta_i) d\theta_i \wedge d\phi_i.$$

The paper is organized as follows. In Sec. II, we introduce all the concepts used in the derivation of the vortex equations. Hally's equations are written as an invariant Hamiltonian system and, for the spherical case, are shown to be equivalent to Bogolmonov's equations. In Sec. III, we derive the expression for the first order perturbation term. The main idea is to write the conformal transformation from the ellipsoid into the plane as a formal series in the eccentricity of the generating ellipse of the ellipsoid. In Sec. IV, we exhibit two simple applications of Eq. (2). The case $N=2$, despite being integrable, seems not to admit explicit solutions for a general initial condition. We characterize the relative equilibria when $|\Gamma_1| = |\Gamma_2|$. For the $N=3$ case, we show numerically, by choosing specific parameters, that the three-vortex problem (2) on the ellipsoid is chaotic.

II. THE EQUATIONS OF HALLY

In this section, we discuss Eq. (1) from a geometric and an invariant point of view. Hally's equations were deduced under the topological constraint that the sum of all vorticities is zero and, in fact, this is essential in recasting the equations in Hamiltonian form. We also show that, under this zero vorticity constraint, Hally's equations on the sphere coincide with Bogolmonov's equations.

A. Hally's equations as a Hamiltonian system on the conformal plane

Let M be a two-dimensional Riemannian manifold with metric tensor g. M is a symplectic manifold with symplectic form w given by the area form induced by g. Let $H: M \to \mathbb{R}$ be the Hamiltonian function. Hamilton's equations are[9]

$$w(X_H, \cdot) = dH, \tag{3}$$

where X_H is the Hamiltonian vector field induced by H. Now we assume that M is conformal to the plane, that is, that there are coordinates $x_i: M \to \mathbb{R}^2$, $i=1,2$, where the metric is given by

$$g(x_1, x_2) = h^2(x_1, x_2)(dx_1 \otimes dx_1 + dx_2 \otimes dx_2)$$

for some function h. In these coordinates, the symplectic (area) form is given by

$$w = h^2(x_1, x_2) dx_1 \wedge dx_2.$$

We introduce complex coordinates $(\xi, \bar{\xi})$ on the plane (x_1, x_2) by

$$\xi = \tfrac{1}{\sqrt{2}}(x_1 + i x_2),$$

$$\bar{\xi} = \tfrac{1}{\sqrt{2}}(x_1 - ix_2).$$

Using $(\xi, \bar{\xi})$, we obtain

$$g(\xi,\bar{\xi}) = h^2(\xi,\bar{\xi})(d\xi \otimes d\xi + d\bar{\xi} \otimes d\bar{\xi})$$

and

$$w = -ih^2(\xi,\bar{\xi})d\xi \wedge d\bar{\xi}.$$

Hamilton's equations (3) become

$$h^2(\xi,\bar{\xi})d\xi \wedge d\bar{\xi}(X_H, \cdot) = d(iH).$$

For convenience, we think of $h^2(\xi,\bar{\xi})d\xi \wedge d\bar{\xi}$ as the new area form and iH as the new Hamiltonian function. Writing $X_H = \dot{\xi}(\partial/\partial\xi) + \dot{\bar{\xi}}(\partial/\partial\bar{\xi})$, the equations become

$$h^2(\xi,\bar{\xi})\dot{\bar{\xi}} = -i\frac{\partial H}{\partial \xi}. \tag{4}$$

We now show that Hally's equations can be written as a Hamiltonian system on the conformal plane. The equations are

$$\bar{\xi}_n' = h^{-2}(\xi_n,\bar{\xi}_n)\left(\sum_k{}' -i\frac{\Gamma_k}{\xi_n - \xi_k} + i\Gamma_n\frac{\partial}{\partial \xi_n}\log(h(\xi_n,\bar{\xi}_n))\right). \tag{5}$$

Multiplying each term of the sum in the right-hand side by $(\bar{\xi}_n - \bar{\xi}_k)/(\bar{\xi}_n - \bar{\xi}_k)$ and adding the zero $i\Gamma_n(\partial/\partial\xi_n)\log(h(\xi_k,\bar{\xi}_k))$ to the right-hand side, we obtain

$$\bar{\xi}_n' = h^{-2}(\xi_n,\bar{\xi}_n)\left(\sum_k{}' -i\Gamma_k\frac{\partial}{\partial \xi_n}\log|\xi_n - \xi_k|^2 + i\Gamma_n\frac{\partial}{\partial \xi_n}\log(h(\xi_n,\bar{\xi}_n)h(\xi_k,\bar{\xi}_k))\right).$$

Assuming that the sum of the vorticities is equal to zero, we have

$$h^2(\xi_n,\bar{\xi}_n)\bar{\xi}_n' = -i\left(\sum_\xi{}'\Gamma_k\frac{\partial}{\partial \xi_n}\log|\xi_n - \xi_k|^2 + \sum_k{}'\Gamma_k\frac{\partial}{\partial \xi_n}\log(h(\xi_n,\bar{\xi}_n)h(\xi_k,\bar{\xi}_k))\right),$$

that is,

$$h^2(\xi_n,\bar{\xi}_n)\bar{\xi}_n' = -i\sum_k{}'\Gamma_k\frac{\partial}{\partial \xi_n}\log(h(\xi_n,\bar{\xi}_n)h(\xi_k,\bar{\xi}_k)|\xi_n - \xi_k|^2). \tag{6}$$

However, those are Hamilton's equations:

$$w(X_H, \cdot) = d(iH), \tag{7}$$

for the symplectic 2-form

$$\tilde{w} = \sum_{i=1}^{N}\Gamma_i h^2(\xi_i,\bar{\xi}_i)d\xi_i \wedge d\bar{\xi}_i \tag{8}$$

and Hamiltonian function

$$H = \frac{1}{2}\sum_{k,n}{}'\Gamma_k\Gamma_n\log(h(\xi_n,\bar{\xi}_n)h(\xi_k,\bar{\xi}_k)|\xi_n - \xi_k|^2). \tag{9}$$

B. From Hally's to Bogolmonov's equations

We denote by $S^2 \subset \mathbb{R}^3$ the two-dimensional sphere

$$S^2 = \{(x,y,z) \in \mathbb{R}^3 | x^2 + y^2 + z^2 = R^2\}.$$

S^2 is a Riemannian manifold with the metric induced by the Euclidean metric in $\mathbb{R}^3$. We introduce spherical coordinates on S^2 through the relations

$$x = R\cos(\phi)\sin(\theta),$$

$$y = R\sin(\phi)\sin(\theta),$$

$$z = R\cos(\theta),$$

with $0 \leqslant \phi < 2\pi$, $0 < \theta < \pi$.

Let us consider the conformal map $\sigma : S^2 \to \mathbb{C}$, $\sigma(\theta,\phi) = u + iv$, where

$$u = \tan(\theta/2)\cos(\phi), \tag{10}$$

$$v = \tan(\theta/2)\sin(\phi).$$

The conformal factor is given by

$$h(\xi,\bar{\xi}) = \frac{2R}{1+\xi\bar{\xi}},$$

where $\xi = u + iv$. Now, let $\mathbf{r}_1, \mathbf{r}_2 \in S^2 \subset \mathbb{R}^3$. The Euclidean distance $\|\mathbf{r}_1 - \mathbf{r}_2\|^2$ in $\mathbb{R}^3$ can be computed in the conformal variables. In fact,

$$\begin{aligned}\|\mathbf{r}_1 - \mathbf{r}_2\|^2 &= (x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2\\ &= R^2\{2 - 2\cos(\theta_1)\cos(\theta_2) - 2\sin(\theta_1)\sin(\theta_2)\cos(\phi_1 - \phi_2)\},\\ &= 4\cos(\theta_1)^2\cos(\theta_2)^2R^2\{(u_1^2 + v_1^2) + (u_1^2 + v_1^2) - 2u_1v_1 - 2u_2v_2\},\\ &= \frac{2R}{1+\xi_1\bar{\xi}_1}\frac{2R}{1+\xi_2\bar{\xi}_2}|\xi_1 - \xi_2|^2,\end{aligned}$$

that is,

$$\|\mathbf{r}_1 - \mathbf{r}_2\|^2 = h(\xi_1,\bar{\xi}_1)h(\xi_2,\bar{\xi}_2)|\xi_1 - \xi_2|^2.$$

Now, let us consider the map $F : S^2 \times \ldots \times S^2 \to \mathbb{C}^N$ given by

$$F(\mathbf{r}_1,\mathbf{r}_2, \ldots ,\mathbf{r}_N) = (\sigma(\mathbf{r}_1),\sigma(\mathbf{r}_2), \ldots ,\sigma(\mathbf{r}_N)).$$

The pull back by F of the symplectic form

$$w = \sum_{i=1}^{N} \Gamma_i h^2(\xi_i,\bar{\xi}_i) d\xi_i \wedge d\bar{\xi}_i$$

is just

$$F^*\tilde{w} = \sum_{i=1}^{N} \Gamma_i w_i,$$

where by w_i we denote the standard area form

$$w_i = x_i dy_i \wedge dz_i + y_i dz_i \wedge dx_i + z_i dx_i \wedge dy_i \tag{11}$$

induced by the Euclidean metric on S_i^2. The pull back of the Hamiltonian function

$$H = \frac{1}{2}\sum_{k,n}{}' \Gamma_k \Gamma_n \log(h(\xi_n, \bar{\xi}_n) h(\xi_k, \bar{\xi}_k) \|\xi_n - \xi_k\|^2)$$

is

$$F^* H = \frac{1}{2}\sum_{k,n}{}' \Gamma_k \Gamma_n \log(\|\mathbf{r}_k - \mathbf{r}_n\|^2). \tag{12}$$

Hamilton's equations for Hamiltonian function (12) and symplectic form (11) are precisely Bogolmonov's equations for N interacting vortices on a sphere.

We notice that Hally's equations are equivalent to Bogolmonov's equations only for the case where all the vorticities sums up to zero. Therefore, in dealing with the general vorticity case, one must use Bogomolnov's equations.

III. EQUATIONS ON THE SYMMETRIC ELLIPSOID–PERTURBATION APPROACH

Let $E^2 \subset \mathbb{R}^3$ represent the two-dimensional ellipsoid of revolution

$$\frac{x^2}{R^2} + \frac{y^2}{R^2} + \frac{z^2}{R^2(1+\epsilon)} = 1, \tag{13}$$

where $R > 0$ and $|\epsilon| \ll 1$. If $\epsilon > 0$, the ellipsoid is prolate. In this case, the eccentricity e of the generating ellipse is given by $e = \sqrt{\epsilon/(1+\epsilon)}$. If $\epsilon < 0$, the ellipsoid is oblate and the eccentricity of the generating ellipse is given by $e = \sqrt{-\epsilon}$. We observe that $\epsilon = \mathcal{O}(e^2)$ and therefore, the truncated first order system is expected to approximate the real system for quite high values of the eccentricity.

We introduce coordinates on E^2 through the relations

$$x = R\cos(\phi)\sin(\theta),$$

$$y = R\sin(\phi)\sin(\theta),$$

$$z = R\sqrt{1+\epsilon}\cos(\theta), \tag{14}$$

with $0 \leq \phi < 2\pi$, $0 < \theta < \pi$. If $\epsilon = 0$, the surface is a sphere of radius R. In this case, the projection

$$u = \tan(\theta/2)\cos(\phi),$$

$$v = \tan(\theta/2)\sin(\phi) \tag{15}$$

is a conformal transformation. We want to find a conformal transformation from E^2 to the conformal plane when $\epsilon \neq 0$. By the symmetry of the ellipsoid, the conformal factor must depend only on the radial variable r and not on the angular variable ψ. We look for a transformation of the form

$$\theta = 2\arctan(r) + 2\sum_{i=1}^{\infty} f_i(r)\epsilon^i,$$

$$\phi = \psi. \tag{16}$$

We observe that when $\epsilon = 0$, the inverse of the above transformation is just transformation (15) written in polar coordinates.

The symmetry of the ellipsoid implies that the equator is a circle of radius R. In the (θ,ϕ) coordinates, the equator has parametrization given by $\theta=\pi/2$ and $\phi\in[0,2\pi)$. Therefore, in the conformal coordinates (u,v), the equator becomes the circle of radius $r=u^2+v^2=1$. We remark that the conformal factor for the ellipsoid, when restricted to the circle of radius 1, is just the usual one for the sphere since the ellipsoid E^2 and the S^2 coincide at the equator. This is of utmost importance for what follows. In fact, since we will impose conformality at each order, this trivial observation implies that $f_i(1)=0$ for all i. This is the condition that will guarantee the uniqueness of our series expansion.

Remark 3.1: Alternatively (and equivalently), the uniqueness of the expansion can be obtained by imposing that the symmetry condition $h(r)=(1/r)h(r^{-1})$ has to be satisfied at all orders.

The algorithmic determination of the $f_i(r)$ is obtained by imposing conformality at each order and follows the standard procedures of classical perturbation theory: Truncate the series at order n, find the conditions to be satisfied, and then assume those conditions to find the term of order $n+1$. To find $f_1(r)$, we impose that

$$\theta = 2\arctan(r) + 2f_1(r)\epsilon,$$

$$\phi = \psi$$

generate a conformal transformation up to first order in ϵ.

The line element for the ellipsoid is

$$ds^2 = R^2\sin^2(\theta)d\phi^2 + R^2(1+\epsilon\sin^2(\theta))d\theta^2.$$

We want to write ds^2 in terms of r and ϕ. To this goal, it suffices to write $\sin(\theta)$ and $d\theta$ in terms of r. However,

$$\begin{aligned}\sin(\theta) &= \sin(2\arctan(r) + 2f_1(r)\epsilon)\\ &= \sin(2\arctan(r)) + 2\cos(2\arctan(r))f_1(r)\epsilon + \mathcal{O}(\epsilon^2)\\ &= 2\sin(\arctan(r))\cos(\arctan(r)) + 2(\cos(\arctan(r))^2 - \sin(\arctan(r))^2)f_1(r)\epsilon + \mathcal{O}(\epsilon^2)\\ &= 2\frac{r}{1+r^2} + 2\frac{(1-r^2)}{1+r^2}f_1(r)\epsilon + \mathcal{O}(\epsilon^2).\end{aligned}$$

This gives

$$\sin(\theta)^2 = \frac{4r^2}{(1+r^2)^2} + \frac{8r(1-r^2)}{(1+r^2)^2}f_1(r)\epsilon + \mathcal{O}(\epsilon^2).$$

Also,

$$d\theta = \left(2\frac{1}{1+r^2} + 2\frac{\partial f_1(r)}{\partial r}\epsilon + \mathcal{O}(\epsilon^2)\right)dr.$$

Therefore,

$$d\theta^2 = \left(4\frac{1}{(1+r^2)^2} + \frac{8}{1+r^2}\frac{\partial f_1(r)}{\partial r}\epsilon + \mathcal{O}(\epsilon^2)\right)dr^2.$$

Finally,

$$ds^2 = \frac{4R^2}{(1+r^2)^2}\left\{\left[1+\frac{2(1-r^2)}{r}f_1(r)\epsilon+\mathcal{O}(\epsilon^2)\right]r^2 d\psi^2 + \left[1+\left(\frac{4r^2}{(1+r^2)^2}+2(1+r^2)\frac{\partial f_1(r)}{\partial r}\right)\epsilon+\mathcal{O}(\epsilon^2)\right]dr^2\right\}.$$

This will be conformal to the plane if it is a multiple of $r^2 d\psi^2 + dr^2$. We observe that if $\epsilon=0$, then the transformation is conformal with conformal factor $\sqrt{4R^2/(1+r^2)^2}$. For $\epsilon\neq 0$, the only way this transformation can be conformal for all values of ϵ is if the coefficient of ϵ in the first square brackets is equal to the coefficient of ϵ in the second square bracket, that is, if

$$(1+r^2)\frac{\partial f_1(r)}{\partial r} - \frac{(1-r^2)}{r}f_1(r) = -\frac{2r^2}{(1+r^2)^2}.$$

This is a linear differential equation for $f_1(r)$. Its general solution (for $r>0$) is equal to

$$f_1(r) = -\frac{\left(\frac{1}{1+r^2}+c\right)}{(1+r^2)}r. \tag{17}$$

Now, by Remark 3.1, we must have $f_1(1)=0$, which implies $c=-\frac{1}{2}$. Finally, we have

$$f_1(r) = \frac{1-r^2}{(1+r^2)^2}r.$$

Transformation (16) becomes

$$\theta = 2\arctan(r) + \frac{(1-r^2)}{(1+r^2)^2}r\epsilon + \sum_{i=2}^{\infty} f_i(r)\epsilon^i,$$

$$\phi = \psi. \tag{18}$$

The conformal factor is, to first order, given by

$$h(r)^2 = \frac{4R^2}{(1+r^2)^2}\left\{1+2\frac{(1-r^2)^2}{(1+r^2)^2}\epsilon\right\},$$

that is,

$$h(r) = \frac{2R}{(1+r^2)}\left\{1+\frac{(1-r^2)^2}{(1+r^2)^2}\epsilon\right\}.$$

We write

$$h(r) = h_0(r) + h_1(r)\epsilon + \mathcal{O}(\epsilon^2), \tag{19}$$

where

$$h_0(r) = \frac{2R}{(1+r^2)} \text{ and } h_1(r) = 2R\frac{(1-r^2)^2}{(1+r^2)^3}.$$

For future reference, we observe that in terms of the variables θ and ϕ, we obtain

$$h_0(r) = 2R\cos^2(\theta/2) \text{ and } h_1(r) = 4R\cos(\theta)\cos^2(\theta/2).$$

Remark 3.2: The above expansion cannot be carried explicitly further. In fact, the equation for the second order factor will be

$$\frac{df_2}{dr} + \frac{1}{2}\frac{r(r^2-1)}{r^2+1}f_2 + \frac{1-13r^2+46r^4-r^{10}-26r^6+9r^8}{4(1+r^2)^5} = 0.$$

The general solution for this equation is given by

$$f_2(r) = \frac{\sqrt{1+r^2}}{4e^{r^2/4}}\left(\int \frac{((r^2-1)^2+4r^2)(r^6-3r^4+7r^2-1)}{(1+r^2)^{11/2}}e^{1/4r^2}dr + 4c\right),$$

where c is an arbitrary constant. The above integral cannot be computed using elementary functions.

A. *N*-vortex equations

Using the conformal factor just found, we will write the N-vortex equations on the ellipsoid of revolution up to first order terms in ϵ. We use the coordinates (θ,ϕ) introduced through Eq. (14). The symplectic 2-form becomes

$$w = \sqrt{1+\epsilon}\, d\phi \wedge d(\cos(\theta)).$$

Observing that

$$\sqrt{1+\epsilon}\, d\phi \wedge d(\cos(\theta))(X_H,\cdot) = d\phi \wedge d(\cos(\theta))(\sqrt{1+\epsilon}X_H,\cdot),$$

we define the vector field $\tilde{X}=\sqrt{1+\epsilon}X_H$. $\tilde{X}$ satisfies Hamilton's equations

$$d\phi \wedge d(\cos(\theta))(\tilde{X},\cdot) = dH, \tag{20}$$

and the flow $\phi_s(x)$ of the vector field $\tilde{X}_h$ is a time reparametrization of the flow $\phi_t(x)$ induced by X_H. In fact, $t=\sqrt{1+\epsilon}s$. We therefore consider Eq. (20) with H and h given by (9) and (19), respectively. H is given by

$$H = \frac{1}{2}\sum_{i,j}{}' \Gamma_i\Gamma_j \ln[(h_0(r_i)+\epsilon h_1(r_i))(h_0(r_j)+\epsilon h_1(r_j))|\xi_i-\xi_j|^2 + \mathcal{O}(\epsilon^2)],$$

that is,

$$H = \frac{1}{2}\sum_{i,j}{}' \Gamma_i\Gamma_j\left[\ln(h_0(r_i)h_0(r_j)|\xi_i-\xi_j|^2) + \epsilon\left(\frac{h_1(r_i)}{h_0(r_i)} + \frac{h_1(r_j)}{h_0(r_j)}\right)\right] + \mathcal{O}(\epsilon^2).$$

The Hamiltonian function H is a function on the variables (θ,ϕ). Moreover, the symplectic 2-form (20) is the standard symplectic form over the sphere. Therefore, we write

$$H = \frac{1}{2}\sum_{i,j}{}' \Gamma_i\Gamma_j[\ln(2-2\cos(\theta_i)\cos(\theta_j)-2\sin(\theta_i)\sin(\theta_j)\cos(\phi_i-\phi_j)) + \epsilon(\cos(\theta_i)^2+\cos(\theta_j)^2)] + \mathcal{O}(\epsilon^2).$$

We conclude that, to first order in ϵ, the dynamics of the N vortices on the ellipsoid is determined by the Hamiltonian system with Hamiltonian function

$$H = \frac{1}{2}\sum_{i,j}{}' \Gamma_i\Gamma_j[\ln(2-2\cos(\theta_i)\cos(\theta_j)-2\sin(\theta_i)\sin(\theta_j)\cos(\phi_i-\phi_j)) + \epsilon(\cos(\theta_i)^2+\cos(\theta_j)^2)]$$

and symplectic 2-form

$$w = \sum_{i=1}^{N} \Gamma_i d\phi_i \wedge d(\cos(\theta_i)).$$

Alternatively, in Cartesian coordinates, the above Hamiltonian system becomes

$$H = \frac{1}{2}\sum_{i,j}{}' \Gamma_i \Gamma_j \log \|\mathbf{r}_i - \mathbf{r}_j\|^2 + \frac{\epsilon}{2}\sum_{i,j}{}' \Gamma_i \Gamma_j (z_i^2 + z_j^2), \tag{21}$$

with symplectic 2-form given by

$$w = \sum_{i=1}^{n} \Gamma_i (x_i dy_i \wedge dz_i + y_i dz_i \wedge dx_i + z_i dx_i \wedge dy_i). \tag{22}$$

IV. APPLICATIONS: THE N=2,3 CASES

A. Two-vortex problem

When $N=2$, the Hamiltonian system given by (21) and (22) reduces to

$$\dot{\mathbf{r}}_1 = \Gamma_2 \frac{\mathbf{r}_2 \times \mathbf{r}_1}{\|\mathbf{r}_1 - \mathbf{r}_2\|^2} + \epsilon \Gamma_2 \mathbf{r}_1 \times z_1 \hat{z},$$

$$\dot{\mathbf{r}}_2 = \Gamma_1 \frac{\mathbf{r}_1 \times \mathbf{r}_2}{\|\mathbf{r}_1 - \mathbf{r}_2\|^2} + \epsilon \Gamma_1 \mathbf{r}_2 \times z_2 \hat{z}. \tag{23}$$

Doing the following change of variables:

$$\mathbf{w} = \Gamma_1 \mathbf{r}_1 - \Gamma_2 \mathbf{r}_2,$$

$$\mathbf{c} = \Gamma_1 \mathbf{r}_1 + \Gamma_2 \mathbf{r}_2, \tag{24}$$

we obtain

$$\dot{\mathbf{w}} = \frac{\mathbf{c} \times \mathbf{w}}{\|\alpha_1 \mathbf{c} - \alpha_2 \mathbf{w}\|^2} + \epsilon(\Delta_1(c_z, w_z)\mathbf{c} \times \hat{z} + \Delta_2(c_z, w_z)\mathbf{w} \times \hat{z}), \tag{25}$$

$$\dot{\mathbf{c}} = \epsilon(\Delta_2(c_z, w_z)\mathbf{c} \times \hat{z} + \Delta_1(c_z, w_z)\mathbf{w} \times \hat{z}),$$

where

$$\alpha_1 \equiv \frac{1}{2}\frac{\Gamma_2 - \Gamma_1}{\Gamma_1 \Gamma_2}, \quad \alpha_2 \equiv \frac{1}{2}\frac{\Gamma_1 + \Gamma_2}{\Gamma_1 \Gamma_2},$$

$$\Delta_1 \equiv \frac{c_z + w_z}{4\Gamma_1^2} + \frac{c_z - w_z}{4\Gamma_2^2} \text{ and } \Delta_2 \equiv \frac{c_z + w_z}{4\Gamma_1^2} - \frac{c_z - w_z}{4\Gamma_2^2}.$$

Equation (25) is singular when $\alpha_1 \mathbf{c} = \alpha_2 \mathbf{w}$, i.e., $\mathbf{r}_1 = \mathbf{r}_2$. When $\epsilon = 0$, Eq. (25) reduces to a linear system and can be easily solved. Those are the well known solutions for the two-vortex problem over the sphere: In a system of coordinates such that its z axis is in the same direction as the center of vorticity $\mathbf{c}$, the solutions are given in the $\mathbf{r}_1$, $\mathbf{r}_2$ variables by

$$\mathbf{r}_1(t) = \frac{1}{2\Gamma_1}(\mu \cos(\theta), \mu \sin(\theta), \|\mathbf{c}\| + w_z),$$

$$\mathbf{r}_2(t) = \frac{1}{2\Gamma_2}(-\mu\cos(\theta), -\mu\sin(\theta), \|\mathbf{c}\| - w_z), \tag{26}$$

where

$$\theta \equiv \gamma t + \phi,$$

with ϕ a constant and

$$\gamma = \frac{\|\mathbf{c}\|}{\alpha_2^2\mu^2 + (\alpha_2 w_z - \alpha_1\|\mathbf{c}\|)^2}, \tag{27}$$

with

$$\mu^2 = w_x^2 + w_y^2 = w^2 - w_z^2 \text{ and } \mu \geqslant 0.$$

We observe that all solutions are periodic and the relative distance between the vortices is a first integral.

When $\epsilon \neq 0$, the center of vorticity vector $\mathbf{c}$ is not preserved. In fact, only its z component c_z is preserved. This first integral implies, by dimensionality, that the system is integrable. Since $\|\mathbf{r}_1\| = \|\mathbf{r}_2\| = 1$, we obtain that

$$\|\mathbf{c}\|^2 + \|\mathbf{w}\|^2 = k_1 \text{ and } \mathbf{c}\cdot\mathbf{w} = k_2$$

are constants of motion.

The relative equilibria can be easily characterized in the cases $|\Gamma_1| = |\Gamma_2| \equiv \Gamma$. Then, we obtain that

$$\|\mathbf{c}\|^2 + \|\mathbf{w}\|^2 = \Gamma \text{ and } \mathbf{c}\cdot\mathbf{w} = 0.$$

First, we characterize the solutions such that $\|\mathbf{r}_1(t) - \mathbf{r}_2(t)\|$ is constant along the motion. Since

$$\|\mathbf{r}_1 - \mathbf{r}_2\|^2 = \|\mathbf{c}\|^2\alpha_1^2 + \|\mathbf{w}\|^2\alpha_2^2$$

and since for $|\Gamma_1| = |\Gamma_2|$, either $\alpha_1 = 0$ or $\alpha_2 = 0$, it follows that $\|\mathbf{c}\|^2$ and $\|\mathbf{w}\|^2$ are constants for the relative equilibria. Therefore,

$$\mathbf{c}\cdot\dot{\mathbf{c}} = \epsilon\Delta_2\mathbf{c}\cdot(\mathbf{w}\times\hat{z}) = 0. \tag{28}$$

$\Delta_2 = 0$ implies that w_z is a constant and therefore $\dot{w}_z = 0$. From the equations it follows that $\mathbf{c}\cdot(\mathbf{w}\times\hat{z}) = 0$. This implies that the only way (28) can be zero is if $\mathbf{c}\cdot(\mathbf{w}\times\hat{z}) = 0$. We arrive at the result that in relative equilibria $\|\mathbf{c}\|^2$, $\|\mathbf{w}\|^2$ and w_z are constants and that the vectors $\mathbf{c}$, $\mathbf{w}$, and $\hat{z}$ are always in the same plane. The vectors $\mathbf{c}$ and $\mathbf{w}$ will precess around the z axis with constant angular velocity and keeping their relative orientation. It follows from the definition that $\mathbf{r}_1$ and $\mathbf{r}_2$ will also precess around the z axis with constant angular velocity and keeping their relative orientation.

We compute the angular velocity of the relative equilibria in the cases where $|\Gamma_1| = |\Gamma_2|$.

Case $\Gamma_1 = \Gamma_2$. If $\Gamma_1 = \Gamma_2 = \Gamma$, then $\alpha_1 = 0$ and $\alpha_2 = \Gamma^{-1}$. Assuming that $c_x = A\cos(\Omega t)$, $c_y = A\sin(\Omega t)$, $w_x = B\cos(\Omega t)$, and $w_y = B\sin(\Omega t)$, we obtain that the angular velocity is given by

$$\Omega = \frac{\Gamma^2}{\|\mathbf{w}\|^2}\left(c_z - \frac{\sqrt{\|\mathbf{c}\|^2 - c_z^2}}{\mu}\right) + \frac{\epsilon}{2\Gamma^2}\left(c_z + w_z\frac{\sqrt{\|\mathbf{c}\|^2 - c_z^2}}{\mu}\right).$$

We observe that the perturbation factor in the angular velocity is proportional to $1/\Gamma^2$, and therefore, the deviation from the corresponding spherical relative equilibria should be bigger for smaller vorticities. We also observe that when $\epsilon = 0$, then, taking as before $\|\mathbf{c}\| = |c_z|$, the above expression reduces to (27).

Case $\Gamma_1 = -\Gamma_2$. If $\Gamma_1 = -\Gamma_2 = \Gamma$, then $\alpha_1 = \Gamma^{-1}$ and $\alpha_2 = 0$. Proceeding as before, we obtain

$$\Omega = \frac{\Gamma^2}{\|\mathbf{c}\|^2}\left(c_z - \frac{\sqrt{\|\mathbf{c}\|^2 - c_z^2}}{\mu}\right) + \frac{\epsilon}{2\Gamma^2}\left(c_z + w_z\frac{\sqrt{\|\mathbf{c}\|^2 - c_z^2}}{\mu}\right).$$

The perturbation term is equal in both cases and again, as $\epsilon \to 0$, we will have $w \to \gamma$.

B. The three-vortex problem

The symmetry breaking when $\epsilon \neq 0$ is expected to destroy the integrability of the three-vortex spherical problem (we refer the reader to Refs. 10 and 11 for symmetry breaking perturbations of integrable planar domains). In fact, when $\epsilon=0$, the center of vorticity vector $\mathbf{c}=\Gamma_1\mathbf{r}_1+\Gamma_2\mathbf{r}_2+\Gamma_3\mathbf{r}_3$ is conserved. The integrability follows by dimensionality. When $\epsilon \neq 0$, only its z component $c_z=\Gamma_1 z_1+\Gamma_2 z_2+\Gamma_3 z_3$ is conserved. That the integrability is destroyed is shown numerically in this section. We do not pursue a complete numerical study of the problem. This will be addressed in a future work.

To calculate the Poincaré sections for the three-vortex problem, we will introduce an appropriate coordinate system. First we introduce a cylindrical system of coordinates $(z_1, z_2, z_3, \phi_1, \phi_2, \phi_3)$ with $\phi_i \in [0, 2\pi)$ and $z_i \in (-1, 1)$ for $i=1,2$. The transformation is defined through

$$x_i = \sqrt{1 - z_i^2}\cos(\phi_i),$$

$$y_i = \sqrt{1 - z_i^2}\sin(\phi_i),$$

$$z_i = z_i.$$

Since

$$w = \sum_{i=1}^{3} \Gamma_i x_i dy_i \wedge dz_i + y_i dz_i \wedge dx_i + z_i dx_i \wedge dy_i = \sum_{i=1}^{3} \Gamma_i d\phi_i \wedge dz_i,$$

this is a symplectic transformation and the equations in cylindrical coordinates are given by

$$\dot{\phi}_i = \frac{1}{\Gamma_i}\frac{\partial H}{\partial z_i} \text{ and } \dot{z}_i = -\frac{1}{\Gamma_i}\frac{\partial H}{\partial \phi_i} \quad \text{for } i = 1,2.$$

To reduce the symmetry, we define the coordinates

$$q_1 = \Gamma_1 z_1 - \Gamma_3 z_3,$$

$$q_2 = \Gamma_1 z_1 - \Gamma_2 z_2,$$

$$q_3 = \Gamma_1 z_1 + \Gamma_2 z_2 + \Gamma_3 z_3,$$

$$p_1 = \frac{\phi_1}{3} + \frac{\phi_2}{3} - 2\frac{\phi_3}{3}$$

$$p_2 = \frac{\phi_1}{3} - 2\frac{\phi_2}{3} + \frac{\phi_3}{3},$$

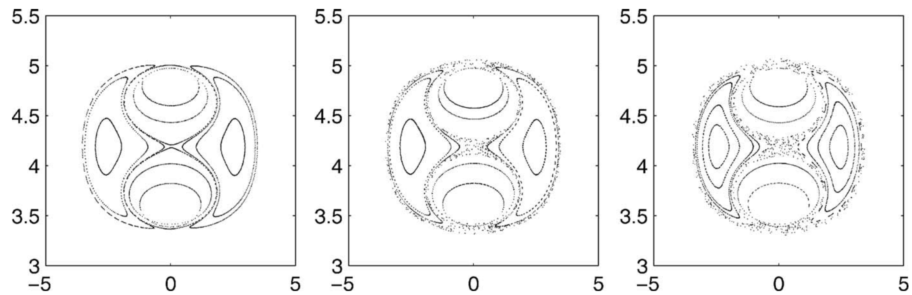

FIG. 1. Poincare maps for the three-vortex problem. Here, $\Gamma_1=\Gamma_2=\Gamma_3=3$, energy$=-\frac{1}{2}$, and $c_z=-\frac{1}{4}$. q_2 is the horizontal axis and p_2 is the vertical axis. The values of ϵ are, from left to right, 0.00, 0.05, 0.10, respectively.

$$p_3 = \frac{\phi_1}{3} + \frac{\phi_2}{3} + \frac{\phi_3}{3}.$$

It follows that

$$dq_1 \wedge dp_1 + dq_2 \wedge dp_2 + dq_3 \wedge dp_3 = \Gamma_1 dz_1 \wedge d\phi_1 + \Gamma_2 dz_2 \wedge d\phi_2 + \Gamma_3 dz_3 \wedge d\phi_3.$$

Therefore, the new system of coordinates forms a canonical system. We observe that $q_3(t) = c_z$, and therefore, $\dot{q}_3(t) = \partial H / \partial p_3 = 0$. Then, p_3 is a cyclic variable and the Hamiltonian (21) is given by

$$H = H(q_1, q_2, p_1, p_2, c_z).$$

Since this Hamiltonian is now defined in a four-dimensional phase space, the Poincare maps can be visualized. The equations are integrated numerically using the Fehlberg-7(8) Runge–Kuta. The relative error in energy is kept smaller than 10^{-10}. A constant step size equal to 10^{-3} is used in all integrations.

Figure 1 shows the pictures of three Poincare maps. The onset of chaotic behavior is clear. It appears (as expected) around the homoclinic solution present in the integrable case $\epsilon=0.00$. We observe that for $\epsilon=0.05$ and $\epsilon=0.10$, the eccentricity of the generating ellipses are 0.22 and 0.33, respectively.

We conclude by raising an open problem: Would it be possible, using a Melnikov-type method, to prove analytically (see, e.g., Ref. 12) the nonintegrability of the truncated perturbed three-vortex system?

ACKNOWLEDGMENTS

The authors thank an anonymous referee for comments and suggestions.

[1] H. Helmholtz, Philos. Mag. **33** (1887).
[2] G. Kirchhoff, *Vorlensungen Uber Matematische Physik*, 3rd ed. (Teubner, Leipzig, 1883).
[3] V. A. Bogolmonov, Fluid Dyn. **6**, 863 (1977).
[4] Y. Kimura and H. Okamoto, J. Phys. Soc. Jpn. **56**, 4203 (1987).
[5] D. Hally, J. Math. Phys. **21**, 211 (1979).
[6] V. A. Bogolmonov, Izv., Acad. Sci., USSR, Atmos. Oceanic Phys. **15**, 18 (1979).
[7] P. K. Newton, *Applied Mathematical Sciences* (Springer-Verlag, New York, 2001), p. 145.
[8] R. Kidambi and P. K. Newton, Physica D **116**, 143 (1998).
[9] J. E. Marsden and T. S. Ratiu, *Introduction to Mechanics and Symmetry*, 2nd ed., Texts in Applied Mathematics Vol. 17 (Springer-Verlag, New York, 1999).
[10] L. Zannetti and P. Franzese, Physica D **76**, 99 (1994).
[11] L. Zannetti and P. Franzese, Eur. J. Mech. B/Fluids **12**, 43 (1993).
[12] J. Koiller and S. Pinto de Carvalho, Commun. Math. Phys. **120**, 643 (1989).